TASMIAH HAQUE

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Professional Summary

AI/ML researcher with 3+ years of experience building real-time intelligent systems at the intersection of video analytics, probabilistic modeling, and edge deployment. *Ph.D. candidate* (*expected Nov 2026*) specializing in generative time-series models, motion keypoint prediction, and bandwidth-efficient inference pipelines. Published author with peer-reviewed works; proven track record in both academic research and industrial settings.

Technical Skills

Machine Learning | Time-series forecasting, probabilistic modeling, generative models (VAE, Normalizing Flows), MCMC, anomaly detection

Computer Vision / Video AI | Motion keypoint prediction, keypoint-based video motion transfer, real-time video synthesis, bandwidth-efficient AI pipelines

Frameworks & Languages | Python, PyTorch, PyTorchTS, TensorFlow, R, MATLAB, C

Infrastructure | NVIDIA RTX / A10 / A100 GPU computing, Linux, Windows

Industrial & Operations | SAP, CPLEX, Arena simulation, Minitab, AutoCAD, SolidWorks

Professional Experience

Graduate Research Assistant — Data Science *Aug 2023 – Present*
West Virginia University • Morgantown, WV

- Developed a VRNN keypoint-prediction framework for **real-time video motion transfer**, enabling future-frame generation with up to $2\times$ bandwidth reduction for video conferencing, VR, and remote monitoring.
- Designed **inference-time stochastic refinement** for a GRU–Normalizing Flow model using MCMC, improving the diversity–fidelity tradeoff in generated video motion without retraining the base model.
- Built and benchmarked **generative time-series models** (VRNN, GRU-NF) in PyTorch/PyTorchTS for accuracy–diversity analysis across video synthesis and industrial anomaly detection tasks.
- Designed a remote **fall-prediction pipeline** integrating task-aligned motion keypoints, future motion prediction, bandwidth reduction, and anomaly classification for healthcare monitoring.

Graduate Teaching Assistant & Researcher *Aug 2021 – May 2023*
West Virginia University • Morgantown, WV

- Developed a physics-infused LSTM for **marine vessel track association** using AIS data, achieving 95% prediction accuracy; published in *Journal of Marine Science and Engineering*.
- Delivered R programming tutorials for Advanced Analysis of Engineering Data, covering statistical methods, visualization, and engineering data interpretation.

Graduate Engineering Trainee — Inbound Logistics *Dec 2019 – Mar 2021*
Unilever Bangladesh Ltd. • Chittagong, Bangladesh

- Managed inventory operations across 400 SKUs, maintaining safety, quality, and production-readiness in a **high-volume manufacturing environment**.
- Optimized **real-time stock levels** in SAP to align packaging-material availability with production plans, reducing operational disruptions.
- Coordinated material flow, storage optimization, and transportation-cost controls; led relabeling initiative to support **on-time-in-full delivery targets**.

Education

Ph.D. in Industrial Engineering *Expected Nov 2026*
West Virginia University, Morgantown, WV

M.S. in Industrial Engineering *May 2023*
West Virginia University, Morgantown, WV

B.Sc. in Industrial & Production Engineering *Apr 2019*
Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh

Publications, Certifications & Awards

- 5 peer-reviewed and preprint publications spanning video motion transfer, marine surveillance, probabilistic time-series modeling, and LSTM post-training quantization. Full list on Google Scholar. Venues: *Multimedia Tools and Applications*; Asilomar Conference; *Journal of Marine Science and Engineering*; Engineering Applications of Neural Networks; IISE Conference.
- **Six Sigma Green Belt Certificate** — IISE | **Supply Chain Director's Award** — Unilever Bangladesh Ltd. (2020)